STM32F446RET6: core connections

Every pin to wire before the project's own peripherals

Pin numbers match the LCSC symbol (U1, LQFP-64).
Use a short wire and the net label on each pin.

Step		Pins	Name	Net label	Connect to
(1) Power and ground	☐	19, 32, 48, 64	VDD	+3V3	100 nF to GND at each pin, plus one 4.7 µF bulk
	☐	1	VBAT	+3V3	100 nF to GND (no backup battery)
	☐	18, 31, 47, 63	VSS	GND	Ground
(2) VDDA and VCAP	☐	13	VDDA/VREF+	VDDA	From +3V3 through a ferrite bead (or 0 Ω); 1 µF + 100 nF to GND
	☐	12	VSSA/VREF-	GND	Ground
	☐	30	VCAP_1	VCAP	4.7 µF ceramic to GND, ESR under 1 Ω. **Never to 3.3 V.**
(3) Reset and boot	☐	7	NRST	NRST	100 nF to GND; SWD header pin 4; optional reset button to GND
	☐	60	BOOT0	BOOT0	10 kΩ to GND (runs from flash)
(4) Crystals	☐	5, 6	PH0 / PH1	OSC_IN OSC_OUT	8 MHz crystal between them; a load cap from each to GND
	☐	3, 4	PC14 / PC15	OSC32_IN OSC32_OUT	32.768 kHz crystal and load caps (optional, for the RTC)
(5) Debug header	☐	46	PA13	SWDIO	SWD header pin 2
	☐	49	PA14	SWCLK	SWD header pin 3
	☐	55	PB3	SWO	SWD header pin 6 (optional)
(6) USB-C	☐	44	PA11	USB_DM	USB-C D− (join both D− pins), through ESD protection
	☐	45	PA12	USB_DP	USB-C D+ (join both D+ pins), through ESD protection
(7) Signs of life	☐	21	PA5	LED	1 kΩ and an LED to GND
	☐	2	PC13	BUTTON	Button to GND; internal pull-up on in firmware
	☐	16	PA2	UART_TX	UART header TX (USART2)
	☐	17	PA3	UART_RX	UART header RX (USART2)

J2: SWD header

1. +3V3
2. SWDIO (PA13)
3. SWCLK (PA14)
4. NRST
5. GND
6. SWO (PB3), optional

J3: UART header

1. UART_TX (PA2)
2. UART_RX (PA3)
3. GND

Board TX goes to the adapter's RX, and RX to TX.

Off the chip: USB-C and power

- CC1 and CC2: each its own 5.1 kΩ to GND
- VBUS to a 3.3 V regulator that makes +3V3
- Regulator caps from its datasheet
- ESD protection on D+, D− and VBUS

Free for the project: 38 pins

8 PC0 **9** PC1 **10** PC2 **11** PC3 **14** PA0 **15** PA1 **20** PA4 **22** PA6 **23** PA7 **24** PC4 **25** PC5 **26** PB0
27 PB1 **28** PB2 **29** PB10 **33** PB12 **34** PB13 **35** PB14 **36** PB15 **37** PC6 **38** PC7 **39** PC8 **40** PC9
41 PA8 **42** PA9 **43** PA10 **50** PA15 **51** PC10 **52** PC11 **53** PC12 **54** PD2 **55** PB3* **56** PB4 **57** PB5
58 PB6 **59** PB7 **61** PB8 **62** PB9

*PB3 is SWO if you use it. Assign these in CubeMX before labelling them.

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EasyEDA Standard shortcuts

Default keys for the editor at easyeda.com

From the official EasyEDA Standard shortcut list:
docs.easyeda.com/en/Introduction/Shortcut-Keys

Everywhere

Shift + F	Search the library
Space or R	Rotate
X	Flip horizontal
Y	Flip vertical
A	Zoom in
Z	Zoom out
K	Fit window
Hold right button	Pan
Esc	Cancel the current tool
Tab	Edit attributes while placing
Delete	Delete selected
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + S	Save
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + F	Find a component
Ctrl + D	Design manager
Shift + X	Cross probe
Alt + W	Close the current tab

Schematic

W	Wire
N	Net Label
Ctrl + Q	VCC net flag
Ctrl + G	GND net flag
B	Bus
U	Bus entry
T	Text
P	Pin (symbol editor)
L	Polyline: a drawing, not a wire
I	Edit the selected symbol
Alt + F	Footprint manager
Shift + T	Symbol wizard

PCB

W	Track
V	Via
P	Pad
T	Top layer
B	Bottom layer
1 – 4	Inner layers
L	Change routing angle
+ or –	Wider or narrower track while routing
*	Next signal layer
Delete	Undo a track segment while routing
H	Highlight a net
M	Measure
Q	Change units
E	Copper area
Shift + B	Rebuild all copper areas
Shift + M	Remove copper fill
Ctrl + L	Layer manager
Ctrl + Q	Show or hide net names

Where is that command?

Find a part	**Shift+F, or Library in the left panel**
Check nets and errors	**Ctrl+D, or Design in the left panel**
Check footprints	**Alt+F (Footprint manager)**
Make the PCB	**Convert to PCB icon on the schematic toolbar**
Send later schematic edits	**Update PCB button, or in the PCB: Design › Import Changes**
Track widths and spacing	**Design › Design Rule...**
Files for the factory	**Fabrication › PCB Fabrication File(Gerber)**
Parts list	**Fabrication › BOM..., or File › Export BOM...**
Order boards	**Save to Cart in the Gerber dialog**

Drawing Tools lines (polyline, L) look like wires but connect nothing. Use Wire (W). Keys can be changed, so a shared computer may differ.

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